



Malaviya National Institute of Technology (MNIT) Jaipur

Project title:

**Decoding Consumer Behavior: A Data-Driven Approach to
Retail Decision Making Using an Automated Analytics Pipeline**

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TABLE OF CONTENTS

1. Introduction & Problem Statement
2. Project Objectives
3. Tools & Technologies Used
4. Step-by-Step Project Workflow
 - 4.1 Excel File Structure
 - 4.2 Python Reads the Excel File
 - 4.3 Python Inserts Data into MySQL
 - 4.4 Power BI Connects to MySQL
 - 4.5 Building the Power BI Dashboard
5. Challenges & Problems Faced
6. Final Output & Results
7. Conclusion

1. INTRODUCTION & PROBLEM STATEMENT

1.1 Project Overview

This project presents a complete end-to-end data analytics pipeline. It extracts raw consumer transaction data stored in Excel, processes it with Python, loads it into a MySQL relational database, and visualizes key business metrics in an interactive Power BI dashboard.

1.2 Problem Statement

A fast-growing retail company operating through both online and offline channels observed inconsistent sales performance, shifting customer preferences, and declining repeat-purchase rates across several product categories. Although the company collected large volumes of customer transaction data, management lacked clear visibility into the factors actually driving purchasing decisions and long-term loyalty.

Specific business challenges:

- Certain products performed well only during specific seasons, but inventory planning was not optimised accordingly.
- Marketing campaigns generated traffic, yet conversion rates remained inconsistent across different customer groups.
- Customers from different age groups and regions exhibited different shopping behaviours, making personalised marketing difficult.
- Discounts and free shipping increased sales temporarily, but the company was unsure whether they improved profitability or customer retention in the long run.
- Online and offline sales channels exhibited different purchasing patterns, but these insights were not being effectively utilised.
- Customer reviews, ratings, and payment preferences potentially influenced buying behaviour, but their impact had not been quantified.

2. PROJECT OBJECTIVES

1. Improve customer retention and loyalty by identifying behavioral patterns associated with repeat purchases.
2. Increase sales and revenue by uncovering high-performing products, regions, and customer segments.
3. Optimize marketing campaigns and discount strategies by quantifying their true impact on profitability and retention.
4. Identify profitable customer segments using demographic and behavioural attributes such as age group, gender, and location.
5. Enhance product and inventory planning by detecting seasonal demand trends and category-level performance.
6. Deliver a more personalised shopping experience by understanding channel-specific purchasing patterns (online vs. offline).
7. Automate the data pipeline from Excel to MySQL using Python, eliminating manual data entry and reporting errors.
8. Build an interactive Power BI dashboard that enables non-technical stakeholders to explore data and derive insights independently.

3. TOOLS & TECHNOLOGIES USED

Tool / Technology	Version / Type	Purpose
Microsoft Excel	.xlsx Format	Raw data storage — customer_shopping_behavior.xlsx
Python	Python 3.14.4	Core scripting for data extraction, transformation, loading
pandas	Python Library	Reading Excel, data cleaning, DataFrame operations
mysql-connector-python	Python Library	Connecting Python to MySQL, executing SQL queries
MySQL	8.0.31.msi	Relational database storing structured consumer data
Power BI Desktop	BI Tool	Dashboard creation and data visualisation

4. STEP-BY-STEP PROJECT WORKFLOW

4.1 Excel File Structure — customer_shopping_behavior.xlsx

	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Category	Purchase Amount	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code	Previous Purchases	Payment Method	Frequency of Purchases
2	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
3	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
4	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
5	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
6	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually
7	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Venmo	Weekly
8	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	49	Cash	Quarterly
9	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	19	Credit Card	Weekly
10	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	Venmo	Annually
11	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes	4	Cash	Quarterly
12	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup	Yes	Yes	26	Bank Transfer	Bi-Weekly
13	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes	Yes	10	Bank Transfer	Fortnightly
14	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes	Yes	37	Venmo	Fortnightly
15	Clothing	51	New Hampshire	M	Violet	Spring	4.7	Yes	Express	Yes	Yes	31	PayPal	Weekly
16	Outerwear	53	New York	L	Teal	Winter	4.7	Yes	Free Shipping	Yes	Yes	34	Debit Card	Weekly
17	Clothing	81	Rhode Island	M	Teal	Winter	2.8	Yes	Store Pickup	Yes	Yes	8	PayPal	Monthly
18	Accessories	36	Alabama	S	Gray	Spring	4.1	Yes	Next Day Air	Yes	Yes	44	Debit Card	Bi-Weekly
19	Clothing	38	Mississippi	XL	Lavender	Winter	4.7	Yes	2-Day Shipping	Yes	Yes	36	Venmo	Quarterly
20	Clothing	48	Montana	S	Black	Summer	4.6	Yes	Free Shipping	Yes	Yes	17	Cash	Weekly

Dataset- F:\Data analytics project\customer_shopping_behavior.csv

4.2 Python Reads the Excel File — load_data.py

```
# Loading the dataset using pandas

import pandas as pd

df = pd.read_csv('customer_shopping_behavior.csv')
```

df.head()

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal

Link-<F:\Data analytics project\Customer Shopping Behavior Analysis.ipynb>

4.3 Python Inserts Data into MySQL

Code for MySQL

```
In [ ]: !pip install pymysql sqlalchemy
```

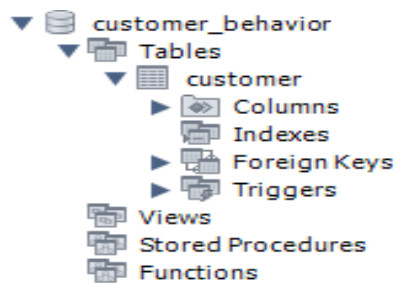
```
In [ ]: from sqlalchemy import create_engine

# MySQL connection
username = "root"
password = "your_password"
host = "localhost"
port = "3306"
database = "customer_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

# Write DataFrame to MySQL
table_name = "customer" # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

# Read back sample
pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)
```



link-<F:\Data analytics project\Customer Shopping Behavior Analysis.ipynb>

4.4 Power BI Connects to MySQL

1. Install MySQL Connector/NET
2. Open Power BI Desktop
3. Get Data → More → Database → MySQL Database
4. Click Connect
5. Enter:
Server: localhost:3306
Database:
customer_shopping_behavior
6. Click ok
7. Enter MySQL Username & Password
8. Click Connect
9. Select tables
10. Click Load
11. MySQL data is successfully loaded

4.5 Building the Power BI Dashboard



Link-<F:\Data analytics project\customer behavior dashboard.pbix>

KPI Overview

Metric	Value	What It Means
Total Customers	3.9K	Dataset covers 3,900 unique customers
Average Purchase Amount	\$59.76	A typical customer spends ~\$60 per transaction
Average Review Rating	3.75 / 5	Customer satisfaction is moderate — room for improvement

Revenue by Category

Clothing leads revenue, followed by Accessories, Footwear, and Outerwear at the bottom. **Action:** Invest in Clothing & Accessories inventory; run promotions on Footwear and Outerwear to lift their numbers.

Revenue by Age Group

Age Group	Revenue	Share
Young Adult	\$62K	26.22%
Middle-Age	\$62K	26.22%
Adult	\$59K	24.98%
Senior	\$56K	24.58%

All four groups contribute nearly equally (~25% each). Young Adults and Middle-Age are tied at the top; Seniors spend the least. **Action:** Target Young Adults via social media and influencer marketing for highest ROI.

Subscription Status

73% of customers are non-subscribers. Only 27% are subscribed. **Action:** Introduce exclusive benefits (discounts, early access) to convert non-subscribers into loyal recurring customers.

Sales by Season

Season	Sales
Fall	60K — Peak
Spring	59K
Winter	59K
Summer	56K — Lowest

Fall is the strongest season. Summer drops by 4K compared to Fall — the only notable dip. **Action:** Maximize stock and marketing in Fall. Run Summer-specific promotions to recover the 4K gap.

Most Preferred Shipping

Free Shipping is the #1 choice across all customers — showing clear price sensitivity. **Action:** Set a minimum order value to qualify for free shipping, pushing average order value higher.

Bottom Insight Cards

Insight	Finding	Business Meaning
Most Revenue	Young Adult (\$62K)	Most valuable customer segment
Subscriber Spending	~1% more than non-subscribers	Subscription program exists but benefits are too weak to drive significant extra spending — needs improvement
Lowest Rated Category	Clothing (3.7 / 5)	Highest revenue category but worst satisfaction — serious long-term risk

The Most Critical Finding — Clothing Contradiction

Clothing earns the **most revenue** but has the **lowest rating (3.7)**. Customers are buying it but not satisfied. Left unaddressed this leads to declining repeats, negative reviews, and eventual revenue loss.

Action: Investigate product quality, sizing accuracy, and delivery condition in the Clothing category immediately.

Overall Business Priorities

Area	Priority
Fix Clothing satisfaction gap	High — urgent
Convert 73% non-subscribers	High
Grow Young Adult & Middle-Age segment	Medium
Summer sales recovery	Medium
Strengthen subscription benefits	Medium
Use Free Shipping as a marketing tool	Low — already working

5. CHALLENGES & PROBLEMS FACED

Challenge 1 — Excel Data Was Messy and Inconsistent

When the Excel file was loaded into pandas, data was not clean. The Gender column had values like "male", "Male", "MALE" — all meaning the same thing but treated as different categories. Similarly Shipping_Type had trailing spaces like "Free Shipping " which created duplicate entries in Power BI slicers and wrong counts.

Solution:

```
python
df['Gender'] = df['Gender'].str.strip().str.title()
df['Shipping_Type'] = df['Shipping_Type'].str.strip()
```

Challenge 2 — Age Group Column Did Not Exist in Raw Data

The **Revenue by Age Group** chart required an Age_Group column (Young Adult, Middle-Age, Adult, Senior) but the original Excel file only had a numeric Age column. There was no ready-made field available for grouping in Power BI.

Solution: Created a calculated column directly in Power BI using DAX:

```
Age_Group =  
SWITCH(  
    TRUE(),  
    shopping_records[Age] >= 18 && shopping_records[Age] <= 25, "Young Adult",  
    shopping_records[Age] >= 26 && shopping_records[Age] <= 35, "Middle-Age",  
    shopping_records[Age] >= 36 && shopping_records[Age] <= 50, "Adult",  
    shopping_records[Age] > 50, "Senior",  
    "Unknown"  
)
```

Challenge 3 — KPI Cards Not Updating When Slicer Was Clicked

After the dashboard was fully built, the slicers for **Gender**, **Category**, and **Subscription Status** were tested. All charts updated correctly when a filter was applied — but the three KPI cards at the top (Total Customers, Avg Purchase Amount, Avg Rating) stayed frozen on the overall numbers and did not respond to slicer selections.

Solution:

```
Total_Customers = DISTINCTCOUNT(shopping_records[Customer_ID])  
Avg_Purchase     = AVERAGE(shopping_records[Purchase_Amount])  
Avg_Rating       = AVERAGE(shopping_records[Review_Rating])
```

FINAL OUTPUT & RESULTS

A fully interactive single-page **Customer Behavior Dashboard** in Power BI connected live to MySQL database . The dashboard reads from the table which was populated by the Python pipeline from `customer_shopping_behavior.xlsx`.

Any time the Excel data is updated and the Python script is re-run, the dashboard reflects the latest numbers automatically on refresh.

CONCLUSION

The dashboard successfully converted 3,900 rows of raw Excel data into 7 clear business findings. The most urgent issue is the **Clothing quality gap** — high revenue but low satisfaction. The biggest growth opportunity is **converting the 73% non-subscribers** into loyal customers. And the easiest revenue win is **setting a free shipping threshold** to push average order value above the current \$59.76.